11. If R is any point on the x-axis and Q is any point on the y-axis and P is a variable point on RQ with $RP = b$, P$Q = a$. then find the equation of locus of P.
12. If the points $P(6, 2)$ and $Q(-2, 1)$ and R are the vertices of a ΔPQR and R is the point on the locus $y = x^2 - 3x + 4$, then find the equation of the locus of centroid of ΔPQR
13. If Q is a point on the locus of $x^2 + y^2 + 4x - 3y + 7 = 0$, then find the equation of locus of P which divides segment OQ externally in the ratio 3:4,where O is origin.
14. Find the points on the locus of points that are 3 units from x-axis and 5 units from the point $(5, 1)$.
15. The sum of the distance of a moving point from the points $(4, 0)$ and $(-4, 0)$ is always 10 units. Find the equation of the locus of the moving point.

6.3 Straight Lines

Linear equations can be rewritten using the laws of elementary algebra into several different forms. These equations are often referred to as the "equations of the straight line."

In the general form the linear equation is written as:

$$ax + by + c = 0 \tag{6.4}$$

where a and b are not both equal to zero. The name "linear" comes from the fact that the set of solutions of such an equation forms a straight line in the plane. In this chapter "line", we mean a straight line unless otherwise stated.

There are many ways to write the equation of a line which can all be converted from one to another by algebraic manipulation. These forms are generally named by the type of information (data) about the line that is needed to write down the form. Some of the important data are **points, slope**, and **intercepts**

6.3.1 The relationship between the angle of inclination and slope

Definition 6.3

The angle of inclination of a straight line is the angle, say θ, made by the line with the x-axis measured in the counter clockwise (positive) direction.

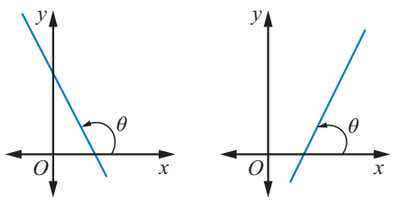


Figure 6.13

Definition 6.4

The slope or gradient of a straight line is a number that measures its "direction and steepness".

The slope of a line in the plane containing the x and y axes, is generally represented by the letter m. It can be measured in many ways as given below:

(i) When θ is the angle of inclination of the line with the x-axis measured in the counter clockwise direction then the slope